

Chronic Kidney Disease (CKD) Prediction Using Machine Learning Algorithms

Objective

The objective of this study was to build and compare multiple machine learning classification algorithms for predicting Chronic Kidney Disease (CKD). The models were evaluated using Accuracy, F1-Score, Cross-Validation Score, ROC-AUC Score, and Confusion Matrix.

Dataset Information

- Problem Type: Binary Classification
- Target Variable:
 - **yes** → CKD Present
 - **no** → CKD Absent
- Total Test Samples: 133

1. Logistic Regression

Best Hyperparameters

```
{'C': 10, 'penalty': 'l2', 'solver': 'lbfgs'}
```

Performance Metrics

Metric	Value
Cross Validation Score	98.88%
Accuracy	98.50%
F1 Macro	98.50%
ROC-AUC	99.98%

Confusion Matrix

Actual / Predicted	Yes	No
Yes	80	2
No	0	51

Observation

- Correctly identified 80 CKD patients.
- Missed only 2 CKD patients.

- No healthy patients were misclassified as CKD.
- Excellent ROC-AUC score indicates strong class separation.

2. Decision Tree Classifier

Best Hyperparameters

```
{'criterion': 'entropy', 'max_features': 'sqrt', 'splitter': 'random'}
```

Performance Metrics

Metric	Value
Cross Validation Score	97.39%
Accuracy	93.98%
F1 Macro	94.04%

Confusion Matrix

Actual / Predicted	Yes	No
Yes	75	7
No	1	50

Observation

- Lowest performance among all models.
- Missed 7 CKD patients.
- Suitable for interpretability but not for highest predictive accuracy.

3. K-Nearest Neighbors (KNN)

Best Hyperparameters

```
{  
  'metric':'manhattan',  
  'n_neighbors':3,  
  'p':1,  
  'weights':'distance'  
}
```

Performance Metrics

Metric	Value
Cross Validation Score	97.40%
Accuracy	96.99%
Weighted F1 Score	97.01%
ROC-AUC	98.76%

Confusion Matrix

Actual / Predicted	Yes	No
Yes	78	4
No	0	51

Observation

- Good performance.
- Zero False Positives.
- Missed 4 CKD patients.
- Performs better than Decision Tree but below Logistic Regression.

4. Naive Bayes

Best Hyperparameters

```
{'var_smoothing':0.005336699231206307}
```

Performance Metrics

Metric	Value
Cross Validation Score	98.50%
Accuracy	97.74%
Weighted F1 Score	97.76%
ROC-AUC	100%

Confusion Matrix

Actual / Predicted	Yes	No
Yes	79	3
No	0	51

Observation

- Excellent ROC-AUC score.

- Very low computational complexity.
 - Missed only 3 CKD patients.
-

5. Random Forest Classifier

Best Hyperparameters

```
{  
  'criterion':'gini',  
  'max_features':'sqrt',  
  'n_estimators':50  
}
```

Performance Metrics

Metric	Value
Cross Validation Score	98.78%
Accuracy	98.50%
F1 Macro	98.50%

Confusion Matrix

Actual / Predicted			Yes	No
Yes		81	1	
No		1		50

Observation

- Strong performance.
 - Only 2 total misclassifications.
 - Robust against overfitting due to ensemble learning.
-

6. Support Vector Machine (SVM)

Best Hyperparameters

```
{  
  'C':10,  
  'degree':2,  
  'gamma':'scale',  
  'kernel':'sigmoid'  
}
```

Performance Metrics

Metric	Value
Cross Validation Score	99.25%
Accuracy	99.25%
F1 Macro	99.25%

Confusion Matrix

Actual / Predicted	Yes	No
Yes	81	1
No	0	51

Observation

- Highest classification performance.
- Only one CKD patient misclassified.
- Zero healthy patients falsely diagnosed as CKD.
- Highest Cross Validation Score among all models.

Comparative Analysis

Model	CV Score	Accuracy	F1 Score	ROC-AUC
Logistic Regression	98.88%	98.50%	98.50%	99.98%
Decision Tree	97.39%	93.98%	94.04%	-
KNN	97.40%	96.99%	97.01%	98.76%
Naive Bayes	98.50%	97.74%	97.76%	100.00%
Random Forest	98.78%	98.50%	98.50%	-
SVM	99.25%	99.25%	99.25%	-

Conclusion

Among all evaluated machine learning algorithms, the **Support Vector Machine (SVM)** achieved the best overall performance with:

- Highest Cross Validation Score (99.25%)
- Highest Accuracy (99.25%)
- Highest F1 Score (99.25%)
- Only one misclassified CKD patient
- Zero False Positives

Therefore, **SVM is the most suitable model for Chronic Kidney Disease prediction on the given dataset**, followed closely by Logistic Regression and Random Forest. Decision Tree

showed the lowest predictive performance and is not recommended when prediction accuracy is the primary objective.

Final Ranking

1. **SVM** ★
2. **Logistic Regression**
3. **Random Forest**
4. **Naive Bayes**
5. **KNN**
6. **Decision Tree**

This comparison demonstrates that kernel-based and ensemble-based methods provide superior performance for CKD classification compared to single-tree and instance-based learning approaches.

Detailed report

OVERALL REPORT for Logistic Regression Classification:

Best Parameters:

```
{'C': 10, 'penalty': 'l2', 'solver': 'lbfgs'}
```

Best Cross Validation Score:

```
0.9888038639876322
```

F1 macro:

```
0.9850141736106648
```

The confusion Matrix:

```
[[80  2]
```

```
 [ 0 51]]
```

ROC_AUC Score:

```
0.9997608799617408
```

classification_report:

	precision	recall	f1-score	support
no	0.96	1.00	0.98	51
yes	1.00	0.98	0.99	82
accuracy		0.98		133
macro avg	0.98	0.99	0.98	133
weighted avg	0.99	0.98	0.99	133

OVERALL REPORT for DecisionTree Classification:

Best Parameters:

```
{'criterion': 'entropy', 'max_features': 'sqrt', 'splitter': 'random'}
```

Best Cross Validation Score:

0.9739047731688905

F1 macro:

0.940378372090564

The confusion Matrix:

```
[[50  1]
```

```
[ 7 75]]
```

classification_report:

	precision	recall	f1-score	support
no	0.88	0.98	0.93	51
yes	0.99	0.91	0.95	82
accuracy		0.94		133
macro avg	0.93	0.95	0.94	133
weighted avg	0.94	0.94	0.94	133

OVERALL REPORT for KNN Classification:

Best Parameters:

```
{'metric': 'manhattan', 'n_neighbors': 3, 'p': 1, 'weights': 'distance'}
```

Best Cross Validation Score:

0.9739513796952457

The f1_weighted value for best parameter:

0.9701163285572423

The confusion Matrix:

```
[[51  0]
```

```
[ 4 78]]
```

ROC_AUC Score:

0.9875657580105213

classification_report:

		precision	recall	f1-score	support
	no	0.93	1.00	0.96	51
	yes	1.00	0.95	0.97	82
	accuracy		0.97		133
	macro avg	0.96	0.98	0.97	133
	weighted avg	0.97	0.97	0.97	133

OVERALL REPORT for NavieBayes Classification:

Best Parameters:

{'var_smoothing': 0.005336699231206307}

Best Cross Validation Score:

0.9850034030491555

The f1_weighted value for best parameter:

0.9775556904684072

The confusion Matrix:

[[51 0]

[3 79]]

ROC_AUC Score:

1.0

classification_report:

		precision	recall	f1-score	support
	no	0.94	1.00	0.97	51
	yes	1.00	0.96	0.98	82
	accuracy		0.98		133
	macro avg	0.97	0.98	0.98	133
	weighted avg	0.98	0.98	0.98	133

OVERALL REPORT for RandomForest Classification:

Best Parameters:

```
{'criterion': 'gini', 'max_features': 'sqrt', 'n_estimators': 50}
```

Best Cross Validation Score:

```
0.9878138961287377
```

F1 macro:

```
0.9849624060150376
```

The confusion Matrix:

```
[[50  1]
```

```
[ 1 81]]
```

classification_report:

	precision	recall	f1-score	support
no	0.98	0.98	0.98	51
yes	0.99	0.99	0.99	82
accuracy		0.98	133	
macro avg	0.98	0.98	0.98	133
weighted avg	0.98	0.98	0.98	133

OVERALL REPORT for SVM Classification:

Best Parameters:

```
{'C': 10, 'degree': 2, 'gamma': 'scale', 'kernel': 'sigmoid'}
```

Best Cross Validation Score:

```
0.9925200207168186
```

F1 macro:

```
0.9924946382275899
```

The confusion Matrix:

```
[[51  0]
```

```
[ 1 81]]
```

classification_report:

	precision	recall	f1-score	support
no	0.98	1.00	0.99	51
yes	1.00	0.99	0.99	82

accuracy		0.99	133	
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133